

SQL PROJECT

Nova Retail Group Database Analysis

Business Intelligence & Data Analytics

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1. Project Overview

Company Background

Nova Retail Group is a growing retail company operating across multiple regions in South Africa. The company sells consumer products through both physical stores and an online platform. Management needs your help analyzing their data to make better business decisions.

Your Role

You are a Junior Data Analyst tasked with analyzing sales data, customer behavior, and product performance. You will write SQL queries to answer specific business questions and provide insights to management.

Learning Objectives

By completing this project, you will demonstrate proficiency in:

- Basic SQL queries (SELECT, WHERE, ORDER BY)
- Aggregate functions (COUNT, SUM, AVG, MIN, MAX)
- JOINS (INNER JOIN, LEFT JOIN, RIGHT JOIN)
- GROUP BY and HAVING clauses
- Subqueries and CTEs (Common Table Expressions)
- Window functions (RANK, ROW_NUMBER, PARTITION BY)
- Date functions and time-based analysis
- Business intelligence and data interpretation

2. Database Schema

Database Name: NovaRetailDB

Table 1: Products

Contains product catalog information (30 products).

Column	Data Type	Description
ProductID (PK)	VARCHAR(10)	Unique product identifier
ProductName	VARCHAR(100)	Product name
Category	VARCHAR(50)	Product category
UnitPrice	DECIMAL(10,2)	Selling price

CostPrice	DECIMAL(10,2)	Cost to acquire product
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Table 2: Customers

Contains customer information (500 customers).

Column	Data Type	Description
CustomerID (PK)	VARCHAR(10)	Unique customer ID
FirstName, LastName	VARCHAR(50)	Customer name
Region	VARCHAR(20)	North, South, East, West
Channel	VARCHAR(20)	Online or Store
JoinDate	DATE	Date joined

Table 3: Sales

Contains transaction data (2,500 records). This is the main fact table.

Column	Data Type	Description
OrderID (PK)	VARCHAR(20)	Unique order ID
OrderDate	DATE	Transaction date
CustomerID (FK)	VARCHAR(10)	Links to Customers
ProductID (FK)	VARCHAR(10)	Links to Products
Quantity, UnitPrice	INT, DECIMAL	Order details
TotalSales, Profit	DECIMAL(10,2)	Financial metrics
Channel	VARCHAR(20)	Online or Store

Table 4: CustomerFeedback

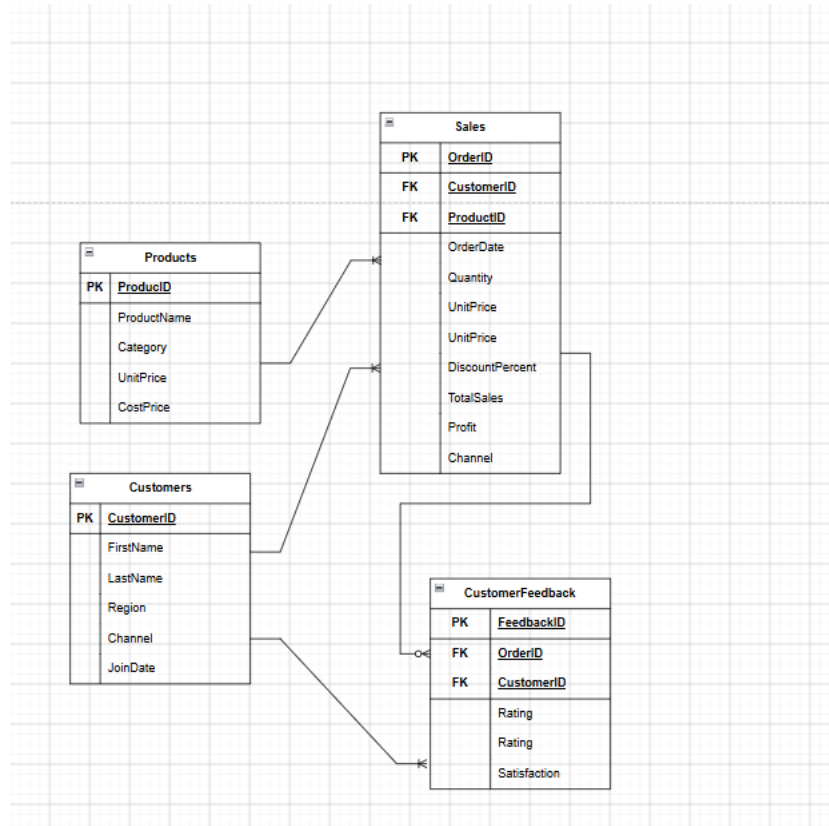
Contains customer satisfaction surveys (1,000 records).

Column	Data Type	Description
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FeedbackID (PK)	VARCHAR(20)	Unique feedback ID
OrderID, CustomerID	VARCHAR	Foreign keys
Rating	INT	1-5 star rating
Satisfaction	VARCHAR(50)	Satisfaction level
RecommendLikelihood	INT	1-10 NPS score

3. Setup Instructions Create an ERD diagram here

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Step 1: Create Database(Ignore)

Run the SQL_Complete_Database_Setup.sql script to create all tables.

Step 2: Import Data

Import the CSV files for Customers, Sales, and CustomerFeedback tables using your database tool.

Step 3: Verify Import

Run the following query to verify all data is loaded:

```

SELECT 'Products' AS TableName, COUNT(*) AS Records FROM Products UNION ALL
SELECT 'Customers', COUNT(*) FROM Customers UNION ALL SELECT 'Sales',
COUNT(*) FROM Sales UNION ALL SELECT 'CustomerFeedback', COUNT(*) FROM
CustomerFeedback;
  
```

Expected Results:

- Products: 30
- Customers: 500
- Sales: 2,500
- CustomerFeedback: 1,000

4. Part 1: Basic SQL Queries (Beginner)

Write SQL queries to answer the following questions. Test each query and paste the results.

Question 1.1: Product Catalog (5 points)

List all products in the Electronics category. Show ProductID, ProductName, and UnitPrice. Order by price from highest to lowest.

Your SQL Query:

```
SELECT productID,  
  
        productName,  
  
        unitPrice  
  
FROM workspace.nova.products  
  
WHERE category = 'Electronics'  
  
ORDER BY unitPrice DESC;
```

Question 1.2: Customer Count (5 points)

How many customers are in each region? Show Region and the count of customers.

Your SQL Query:

```
SELECT region, COUNT(DISTINCT customerID) AS Total_Customers  
  
FROM workspace.nova.customers  
  
GROUP BY region;
```

Question 1.3: Recent Orders (5 points)

Show the 10 most recent orders. Display OrderID, OrderDate, and TotalSales. **Your SQL Query:**

```
SELECT  orderID,

        orderDate,

        TotalSales

FROM workspace.nova.sales

ORDER BY orderDate DESC, orderID DESC

LIMIT 10;
```

Question 1.4: Affordable Products (5 points)

Find all products with a UnitPrice less than R1000. Show ProductName, Category, and UnitPrice.

Your SQL Query:

```
SELECT  ProductName,
        category,
        unitPrice
FROM workspace.nova.products
WHERE unitPrice < 1000;
```

Question 1.5: Customer Satisfaction Summary (5 points)

Count how many feedback responses exist for each Satisfaction level. Order by count descending.

Your SQL Query:

```
SELECT  satisfaction,
        COUNT(feedbackID) AS Number_Of_Feedbacks
FROM workspace.nova.customer_feedback
GROUP BY satisfaction
ORDER BY Number_Of_Feedbacks DESC;
```


5. Part 2: Intermediate SQL Queries

These questions require JOINS, aggregate functions, and GROUP BY clauses.

Question 2.1: Sales by Category (10 points)

Calculate total sales revenue for each product category. Show Category, Total Revenue, and Total Profit. Order by revenue descending.

Your SQL Query:

```
SELECT  p.category,

        ROUND(SUM(s.TotalSales), 2) AS Total_Revenue,

        ROUND(SUM(s.Profit), 2) AS Total_Profit

FROM workspace.nova.products AS p

INNER JOIN workspace.nova.sales AS s

ON p.productID=s.productID

GROUP BY p.category

ORDER BY Total_Revenue DESC;
```

Question 2.2: Top Customers (10 points)

Find the top 5 customers by total purchase amount. Show CustomerID, Customer Name (FirstName + LastName), and Total Spent. Hint: Use CONCAT or + to combine names.

Your SQL Query:

```
SELECT  c.CustomerID,
        CONCAT( c.FirstName, ' ', c.LastName) AS Customer_Name,
        ROUND(SUM(TotalSales), 2) AS Total_Purchase_Amount
FROM workspace.nova.customers AS c
LEFT JOIN workspace.nova.sales AS s
ON c.customerid = s.customerid
GROUP BY CONCAT( c.FirstName, ' ', c.LastName), c.CustomerID
ORDER BY Total_Purchase_Amount DESC
LIMIT 5;
```

Question 2.3: Monthly Sales Trend (10 points)

Show total sales by month for 2024. Display Year, Month (as number), and Total Sales. Order chronologically.

Your SQL Query:

```
SELECT YEAR(OrderDate) AS Year_,  
MONTH(OrderDate) AS Month_Number,  
ROUND(SUM(TotalSales), 2) AS Total_Sales  
FROM workspace.nova.sales  
WHERE YEAR(OrderDate) = 2024  
GROUP BY YEAR(OrderDate), MONTH(OrderDate)  
ORDER BY Year_ ASC, Month_Number ASC;
```

Question 2.4: Channel Performance (10 points)

Compare Online vs Store performance. Show Channel, Number of Orders, Average Order Value, and Total Revenue.

Your SQL Query:

```
SELECT Channel,  
COUNT(OrderID) AS Number_Of_Orders,  
ROUND(AVG(TotalSales)) AS Average_Order_Value,  
ROUND(SUM(TotalSales)) AS Total_Revenue  
FROM workspace.nova.sales  
GROUP BY Channel;
```

Question 2.5: Product Performance with Ratings (10 points)

For each product category, show the average customer rating. Include Category, Average Rating, and Number of Reviews. Only show categories with at least 50 reviews.

Your SQL Query:

```
SELECT p.Category,  
       ROUND(AVG(f.Rating)) AS Average_Rating,  
       COUNT(f.feedbackID) AS Number_of_Reviews  
FROM workspace.nova.customer_feedback AS f  
LEFT JOIN workspace.nova.sales AS s  
ON s.OrderID = f.OrderID  
INNER JOIN workspace.nova.products AS p  
ON s.ProductID = p.ProductID  
GROUP BY p.Category  
HAVING COUNT(f.feedbackID) >= 50;
```

6. Part 3: Advanced SQL Queries

These questions require subqueries, CTEs, window functions, or complex JOINS.

Question 3.1: Best Selling Product per Category (15 points)

Find the product with the highest total sales in each category. Show Category, ProductName, and Total Revenue for that product. Use a window function or subquery.

Your SQL Query:

```
SELECT Category, ProductName, Total_Sales

FROM (

    SELECT  p.Category,

            p.ProductName,

            ROUND(SUM(s.TotalSales), 2) AS Total_Sales,

            RANK() OVER (PARTITION BY p.Category ORDER BY

SUM(s.TotalSales) DESC) AS Product_Rank

    FROM workspace.nova.products AS p

    INNER JOIN workspace.nova.sales AS s

    ON p.ProductID = s.ProductID

    GROUP BY p.Category,

            p.ProductName

) AS ranked_products

WHERE Product_Rank = 1;
```

Question 3.2: Customer Lifetime Value (15 points)

Create a query showing each customer's total purchases, number of orders, and average order value. Also include their region and primary channel. Only show customers with more than 3 orders. Order by total purchases descending. **Your SQL Query:**

```
SELECT c.CustomerID,

       CONCAT(c.FirstName, ' ', c.LastName) AS Customer_Name,

       c.Region,

       c.Channel AS Primary_Channel,

       COUNT(s.OrderID) AS Number_Of_Orders,

       ROUND(SUM(s.TotalSales), 2) AS Total_Purchases,

       ROUND(AVG(s.TotalSales), 2) AS Average_Order_Value

FROM workspace.nova.customers AS c

INNER JOIN workspace.nova.sales AS s

ON c.CustomerID = s.CustomerID

GROUP BY c.CustomerID, c.FirstName, c.LastName, c.Region, c.Channel

HAVING COUNT(s.OrderID) > 3

ORDER BY Total_Purchases DESC;
```

Question 3.3: Profit Margin Analysis (15 points)

Calculate the profit margin percentage for each product. Show ProductName, Category, Total Sales, Total Profit, and Profit Margin %. Order by profit margin descending.

Formula: $(\text{Profit} / \text{Sales}) * 100$ **Your SQL Query:**

```
SELECT p.ProductName, p.Category,

       ROUND(SUM(s.TotalSales), 2) AS Total_Sales,

       ROUND(SUM(s.Profit), 2) AS Total_Profit,

       ROUND((SUM(s.Profit) / SUM(s.TotalSales)) * 100, 2) AS

Profit_Margin_Pct
```

```
FROM workspace.nova.products AS p

INNER JOIN workspace.nova.sales AS s ON p.ProductID = s.ProductID

GROUP BY p.ProductName, p.Category

ORDER BY Profit_Margin_Pct DESC;
```

Question 3.4: Year-over-Year Growth (15 points)

Compare sales between 2023 and 2024. Show the total sales for each year and calculate the growth percentage. Use a CTE or subquery.

Your SQL Query:

```
WITH yearly_sales AS (

    SELECT YEAR(OrderDate) AS Sale_Year, SUM(TotalSales) AS
    Total_Sales

    FROM workspace.nova.sales

    GROUP BY YEAR(OrderDate)

)

SELECT

    (SELECT Total_Sales FROM yearly_sales WHERE Sale_Year = 2023) AS
    Sales_2023,

    (SELECT Total_Sales FROM yearly_sales WHERE Sale_Year = 2024) AS
    Sales_2024,

    ROUND(

        ((SELECT Total_Sales FROM yearly_sales WHERE Sale_Year = 2024) -

        (SELECT Total_Sales FROM yearly_sales WHERE Sale_Year = 2023))

        * 100.0 /

        (SELECT Total_Sales FROM yearly_sales WHERE Sale_Year = 2023),

    2
```

```
) AS Growth_Pct;
```

Question 3.5: Regional Performance Ranking (15 points)

Rank regions by total sales. Use the RANK() or ROW_NUMBER() window function. Show Region, Total Sales, Total Orders, and Rank.

Your SQL Query:

```
SELECT c.Region,  
  
       ROUND(SUM(s.TotalSales), 2) AS Total_Sales,  
  
       COUNT(s.OrderID) AS Total_Orders,  
  
       RANK() OVER (ORDER BY SUM(s.TotalSales) DESC) AS Region_Rank  
  
FROM workspace.nova.customers AS c  
  
INNER JOIN workspace.nova.sales AS s ON c.CustomerID = s.CustomerID  
  
GROUP BY c.Region  
  
ORDER BY Region_Rank;
```


7. Part 4: Business Intelligence Questions

Answer these strategic questions using SQL queries and provide business insights.

Question 4.1: Customer Satisfaction vs. Repeat Purchases (20 points)

Analyze whether highly satisfied customers (rating 4-5) make more repeat purchases than less satisfied customers (rating 1-3). Show satisfaction level groups, average number of orders per customer, and total customers in each group.

Your SQL Query:

```
WITH customer_orders AS (  
    SELECT CustomerID, COUNT(*) AS Num_Orders  
    FROM workspace.nova.sales  
    GROUP BY CustomerID  
)  
SELECT  
    CASE WHEN f.Rating >= 4 THEN 'High Satisfaction (4-5)'  
        ELSE 'Low Satisfaction (1-3)' END AS Satisfaction_Group,  
    ROUND(AVG(co.Num_Orders), 2) AS Avg_Orders_Per_Customer,  
    COUNT(DISTINCT f.CustomerID) AS Total_Customers  
FROM workspace.nova.customer_feedback f  
JOIN customer_orders co ON f.CustomerID = co.CustomerID  
GROUP BY Satisfaction_Group;
```

Business Insight (2-3 sentences):

Happy customers (4-5 star ratings) ordered about the same number of times as unhappy customers (1-3 stars), roughly 6 orders each, so satisfaction barely made a difference. This means how happy someone is doesn't really predict whether they'll keep buying from Nove; people may just be shopping with from Nova out of habit or lack of other options. So management shouldn't assume that making customers happier will automatically lead to more repeat purchases

Question 4.2: Discount Effectiveness (20 points)

Examine the relationship between discount percentage and profit. Create discount bands (0%, 1-10%, 11-20%, 21-30%) and show total sales, total profit, and profit margin for each band.

Your SQL Query:

```
SELECT
    CASE
        WHEN DiscountPercent = 0 THEN '0%'
        WHEN DiscountPercent BETWEEN 1 AND 10 THEN '1-10%'
        WHEN DiscountPercent BETWEEN 11 AND 20 THEN '11-20%'
        WHEN DiscountPercent BETWEEN 21 AND 30 THEN '21-30%'
    END AS Discount_Band,
    ROUND(SUM(TotalSales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(TotalSales)) * 100, 2) AS
Profit_Margin_Pct
FROM workspace.nova.sales
GROUP BY Discount_Band
ORDER BY Discount_Band;
```

Business Insight (2-3 sentences):

The bigger the discount, the less profit we make, margin drops from about 40% with no discount to under 19% at 21-30% off. This means big discounts are mostly just cutting into profit, without clear proof they're bringing in enough extra sales to make up for it. Management should think twice about offering such large discounts unless there's evidence they're actually worth it

Question 4.3: Product Portfolio Optimization (20 points)

Identify underperforming products (bottom 5 by total revenue) and analyze whether they should be discontinued. Consider sales volume, profit margin, and customer ratings.

Your SQL Query:

```
WITH product_totals AS (
    SELECT p.ProductID, p.ProductName, p.Category,
        SUM(s.TotalSales) AS Total_Revenue,
```

```
        SUM(s.Profit) AS Total_Profit,
        ROUND((SUM(s.Profit) * 1.0 / SUM(s.TotalSales)) * 100, 2)
AS Profit_Margin_Pct
FROM workspace.nova.products AS p
JOIN workspace.nova.sales s ON p.ProductID = s.ProductID
GROUP BY p.ProductID, p.ProductName, p.Category
),
product_ratings AS (
    SELECT s.ProductID,
        ROUND(AVG(f.Rating), 2) AS Avg_Rating,
        COUNT(f.FeedbackID) AS Num_Reviews
    FROM workspace.nova.customer_feedback f
    JOIN workspace.nova.sales s ON f.OrderID = s.OrderID
    GROUP BY s.ProductID
)
SELECT pt.ProductName, pt.Category, pt.Total_Revenue,
pt.Profit_Margin_Pct,
    pr.Avg_Rating, pr.Num_Reviews
FROM product_totals pt
LEFT JOIN product_ratings pr ON pt.ProductID = pr.ProductID
ORDER BY pt.Total_Revenue ASC
LIMIT 5;
```

Business Recommendation (3-4 sentences):

The five lowest-selling products are mostly cheap, everyday items, like water bottles, candles, and yoga mats, and their ratings aren't bad, just average. So it doesn't look like people dislike these products; they're probably just low sellers because they're cheap and don't stand out on their own. Instead of cutting them, it might work better to bundle them with bigger, more popular products to boost sales.

8. Part 5: Executive Management Report

Create a comprehensive report for Nova Retail Group's executive team. Your report should include SQL queries and written analysis.

Report Requirements (50 points total)

Section 1: Overall Business Performance (15 points)

Provide a summary of key business metrics:

- Total revenue and profit for 2023-2024
- Overall profit margin
- Total number of orders and customers
- Average order value

Your SQL Query:

```
SELECT
  ROUND(SUM(TotalSales), 2) AS Total_Revenue,
  ROUND(SUM(Profit), 2) AS Total_Profit,
  ROUND((SUM(Profit) / SUM(TotalSales)) * 100, 2) AS
Overall_Profit_Margin_Pct ,
COUNT(OrderID) AS Total_Orders,
COUNT(DISTINCT CustomerID) AS Total_Customers,
ROUND(AVG(TotalSales), 2) AS Average_Order_Value
FROM workspace.nova.sales;
```

	1.2 Total_Revenue	1.2 Total_Profit	1.2 Overall_Profit_Margin_Pct	1.2 Total_Orders	1.2 Total_Customers	1.2 Average_Order_Value
1	16878317.4	5915087.4	35.05	2500	497	6751.33

Section 2: Top 3 Insights (20 points)

Identify and explain the three most important insights from your analysis. Each insight must be supported by SQL query results.

Insight #1: Electronics is the best category by far, it brings in way more money and profit than Home Appliances and Lifestyle Goods put together. Basically, Electronics is carrying the whole business right now.

Supporting SQL Query:

```
SELECT
    CASE
        WHEN DiscountPercent=0 THEN '0%' WHEN DiscountPercent BETWEEN 1
AND 10 THEN '1-10%'
        WHEN DiscountPercent BETWEEN 11 AND 20 THEN '11-20%'
        WHEN DiscountPercent BETWEEN 21 AND 30 THEN '21-30%'
        END AS Discount_Band, ROUND((SUM(Profit)/SUM(TotalSales))*100,2) AS
Margin_Pct
FROM workspace.nova.salesGROUP BY Discount_Band;
```

	Category	1.2 Total Revenue	1.2 Total Profit
1	Electronics	12543376.75	4383976.75
2	Home Appliances	3382900.1	1195540.1
3	Lifestyle Goods	952040.55	335570.55

3 rows 7.015s runtime Refreshed Now

Insight #2: The more discount we give, the less profit we make, profit margin drops from about 40% with no discount, down to under 19% when discounts hit 21-30%. That's a huge drop, so heavy discounting is really costing the business.

Supporting SQL Query:

```
SELECT
    CASE
        WHEN DiscountPercent=0 THEN '0%' WHEN DiscountPercent BETWEEN
1 AND 10 THEN '1-10%'
        WHEN DiscountPercent BETWEEN 11 AND 20 THEN '11-20%'
        WHEN DiscountPercent BETWEEN 21 AND 30 THEN '21-30%'
        END AS Discount_Band,
        ROUND((SUM(Profit)/SUM(TotalSales))*100,2) AS Margin_Pct
FROM workspace.nova.sales
GROUP BY Discount_Band;
```

Table				
	^B _C Discount_Band	1.2 Margin_Pct		
1	0%	39.98		
2	11-20%	27.46		
3	1-10%	35.36		
4	21-30%	18.62		

4 rows 6.107s runtime Refreshed 6m ago

Insight #3: Some regions are doing a lot better than others, the North region brings in about R5 million, while the West region only brings in about R3.8 million. That's a pretty big gap, and it's worth figuring out why West is falling behind.

Supporting SQL Query:

```
SELECT c.Region,
ROUND(SUM(s.TotalSales),2) AS Total_Sales,
RANK() OVER (ORDER BY SUM(s.TotalSales) DESC) AS Region_Rank
FROM workspace.nova.customers c INNER JOIN workspace.nova.sales s
ON c.CustomerID = s.CustomerID
GROUP BY c.Region;
```

Table				
	^B _C Region	1.2 Total_Sales	¹ ₃ Region_Rank	
1	North	4982144.6	1	
2	East	4166999.3	2	
3	South	3895855.75	3	
4	West	3833317.75	4	

4 rows 0.115s runtime Refreshed Now

Section 3: Strategic Recommendations (15 points)

Based on your analysis, provide 3-5 actionable recommendations for management. Each recommendation should be data-driven.

- Keep investing in Electronics:** it's the category making us the most money by far, so we should keep it well-stocked, keep marketing it, and keep adding new products to it.
- Rethink our biggest discounts (21-30% off):** giving that much off is really cutting into our profits. Instead of discounting everything heavily, we should only run big

discounts when we're sure they're actually bringing in enough extra sales to be worth it.

3. **Find out why the West region isn't performing as well:** before spreading our budget equally across all regions, we should look into what North is doing right (staffing, marketing, product choices) that West isn't.
4. **Don't assume happy customers automatically buy more:** our data shows satisfied and unsatisfied customers buy about the same amount, so making customers happier alone won't necessarily bring in more repeat sales. We need other ways to keep people coming back.
5. **Don't discontinue slow selling Lifestyle products yet:** items like water bottles and yoga mats aren't selling much, but their ratings are still decent. Instead of removing them, try bundling them with bestsellers to boost their sales.

9. Submission Guidelines

What to Submit

- This completed document with all SQL queries and answers
- A separate .sql file containing all your queries organized by section
- Screenshots of query results for Part 5 (Management Report)

Grading Criteria

Section	Points	Criteria
Part 1: Basic SQL	25	Correctness, syntax
Part 2: Intermediate SQL	50	JOINS, aggregations
Part 3: Advanced SQL	75	Subqueries, CTEs, windows
Part 4: Business Intelligence	60	Analysis, insights
Part 5: Management Report	50	Strategic thinking
TOTAL	260	

Tips for Success

- Test each query multiple times before submitting
- Use meaningful column aliases to make results clear
- Format your SQL code properly (indentation, line breaks)
- Add comments to complex queries explaining your logic

- Round decimal numbers appropriately (2 decimal places for currency) • Think like a business analyst when interpreting results

Good Luck!